

Shape Derivative-Informed Neural Operators

With Application to Risk-Averse Shape Optimization

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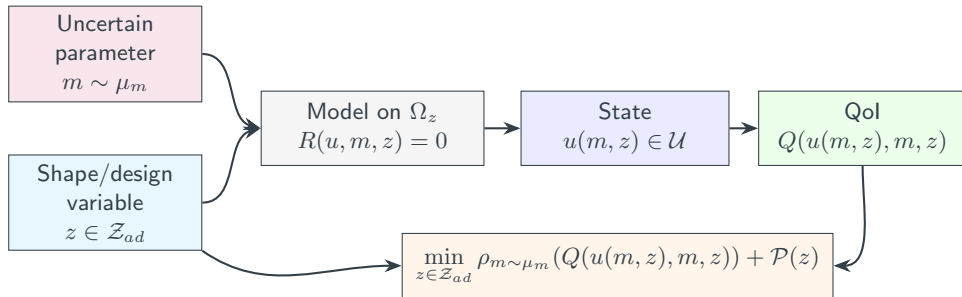
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Motivation: Shape Optimization Under Uncertainty

Shape optimization under uncertainty (OUU) considers PDE models with uncertain parameters $m \sim \mu_m$ and shape variables z . The PDE solution $u(m, z)$ induces a quantity of interest $Q(u(m, z), m, z)$ that depends on both parameter uncertainty and geometric variability.



Instead of optimizing a single realization of Q , OUU seeks shapes that optimize a scalar risk measure $\rho(Q)$, which encodes the designer's risk preference (e.g., expectation, CVaR), together with a regularization term $\mathcal{P}(z)$ on the geometry.

Computational Challenges

Geometric variability

Modeling changes in geometry typically requires re-meshing or mesh deformation, leading to substantial computational overhead for each design variable z .

Sampling complexity

The estimation of risk measures requires repeated forward evaluations and sensitivity evaluations wrt z , resulting in expensive sampling-based optimization:

$$\mathcal{O}(N_{\text{samp}} \times N_{\text{iter}} \times \text{state/sensitivity evaluations}).$$

A surrogate must be fit for the outer-loop task.

- Efficient evaluation of the parametric solution map $(m, z) \in \mathcal{M} \times \mathcal{Z} \mapsto u(m, z)$.
- Provide accurate Fréchet derivatives, especially with respect to z .
- State accuracy alone does not guarantee accurate gradients or stationary points.

Shape-DINO is a **reference-domain** and **derivative-informed** neural operator that enables fast and accurate evaluations for risk-sensitive shape optimization.

Outline

1. Introduction
2. Mathematical Formulation
3. Reduced Basis Neural Operator
4. Theoretical Foundation
5. Numerical Results

PDE on Parametric Spatial Domains

Consider a family of spatial domains $\Omega_z \subset \mathbb{R}^d$, with $d = 2, 3$, parametrized by shape variables $z \in \mathcal{Z} = \mathbb{R}^{d_z}$. Assume each Ω_z is coupled to a reference domain Ω_0 through a diffeomorphism $\chi_z : \mathcal{D} \rightarrow \mathbb{R}^d$ defined by the shape parameters, where $\mathcal{D} \subset \mathbb{R}^d$ is a hold-all domain containing the closure of Ω_0 :

$$\Omega_z = \chi_z(\Omega_0) \subset \mathbb{R}^d, \quad z \in \mathcal{Z} = \mathbb{R}^{d_z}.$$

Moreover, we assume that the Ω_0 is bounded and Lipschitz, such that for each z within some closed admissible set $\mathcal{Z}_{ad} \subset \mathcal{Z}$, the resulting $\Omega_z = \chi_z(\Omega_0)$ is also bounded and Lipschitz.

On each Ω_z , the model parameter and state live on separable Hilbert spaces,

$$\tilde{m} \in \mathcal{M}(\Omega_z), \quad \tilde{u} \in \mathcal{U}(\Omega_z).$$

Then the PDE can be stated abstractly in residual form as

$$\text{given } \Omega_z \text{ and } \tilde{m} \in \mathcal{M}(\Omega_z), \quad \text{find } \tilde{u} \in \mathcal{U}(\Omega_z) \quad \text{such that} \quad \mathcal{R}_z(\tilde{u}, \tilde{m}) = 0, \quad (1)$$

where $\mathcal{R}_z : \mathcal{U}(\Omega_z) \times \mathcal{M}(\Omega_z) \rightarrow \mathcal{V}(\Omega_z)'$ is a differential operator whose output lies in the dual space of some test functions.

PDE on the Reference Domain

Alternatively, the PDE can be equivalently stated on Ω_0 using the pullback through χ_z :

$$u = \tilde{u} \circ \chi_z, \quad m = \tilde{m} \circ \chi_z.$$

Let $\mathcal{U}(\Omega_0), \mathcal{M}(\Omega_0), \mathcal{V}(\Omega_0)$ be the corresponding function spaces defined on Ω_0 . (1) can be equivalently formulated on Ω_0 as

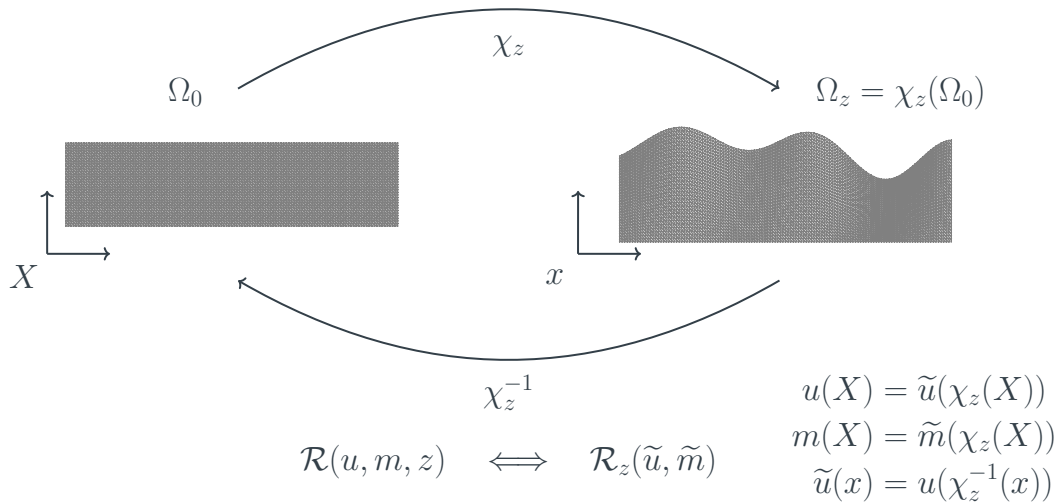
$$\text{given } z \in \mathcal{Z} \text{ and } m \in \mathcal{M}(\Omega_0), \quad \text{find } u \in \mathcal{U}(\Omega_0) \quad \text{such that} \quad \mathcal{R}(u, m, z) = 0, \quad (2)$$

where $\mathcal{R} : \mathcal{U}(\Omega_0) \times \mathcal{M}(\Omega_0) \times \mathcal{Z} \rightarrow \mathcal{V}(\Omega_0)'$ is obtained from $\mathcal{R}_z$ by the appropriate change of variables.

Key point: after pullback, all PDEs are posed on Ω_0 , and the shape variable z appears explicitly in the residual. Under well-posedness, this defines the reference-domain solution operator

$$u = u(m, z), \quad u : \mathcal{M}(\Omega_0) \times \mathcal{Z}_{ad} \rightarrow \mathcal{U}(\Omega_0).$$

Domain Pullback and Variable Mapping



Pullback and Pushforward Operators

For any diffeomorphism $\chi : \Omega_0 \rightarrow \Omega_1$, define the **pullback** operator

$$T_\chi u = u \circ \chi.$$

Its inverse is the **pushforward** operator,

$$T_\chi^{-1} = T_{\chi^{-1}}, \quad T_\chi^{-1} u = u \circ \chi^{-1}.$$

Thus the changes of variables used in (1) and (2) can be expressed as

$$u = T_{\chi_z} \tilde{u}, \quad m = T_{\chi_z} \tilde{m} \quad \text{or} \quad \tilde{u} = T_{\chi_z}^{-1} u, \quad \tilde{m} = T_{\chi_z}^{-1} m,$$

when m and u are scalar fields or componentwise vector-valued fields.

When $\chi_z \in W^{1,\infty}(\Omega_0)$ and $\chi_z^{-1} \in W^{1,\infty}(\Omega_z)$, these composition operators are bounded and linear operators between low order Sobolev spaces on the Ω_z and Ω_0 . Hence the pullback and pushforward give stable changes of variables between Ω_z and Ω_0 (detailed proof in the paper).

Common Coordinate Transformations

Let $x = \chi(X)$, $F_\chi = \nabla_X \chi$, and let $u = T_\chi \tilde{u}$, $v = T_\chi \tilde{v}$ denote componentwise pullbacks. The following transformations hold [1]:

	Spatial form at $x = \chi(X)$	Reference pullback at X
Coordinate	x	$x = \chi(X)$
Scalar field	$\tilde{u}(x) = (T_\chi^{-1} u)(x)$	$u(X) = \tilde{u}(\chi(X)) = (T_\chi \tilde{u})(X)$
Gradient (scalar field)	$\nabla_x \tilde{u}(x)$	$F_\chi^{-T}(X) \nabla_X u(X)$
Gradient (vector field)	$\nabla_x \tilde{v}(x)$	$\nabla_X v(X) F_\chi^{-1}(X)$
Divergence	$\nabla_x \cdot \tilde{v}(x)$	$\text{tr}(\nabla_X v(X) F_\chi^{-1}(X))$
Volume element	dx	$\det F_\chi(X) dX$
Vector surface element	$n(x) ds(x)$	$\det F_\chi(X) F_\chi^{-T}(X) N(X) dS(X)$
Scalar surface element	$ds(x)$	$ \det F_\chi(X) F_\chi^{-T}(X) N(X) dS(X)$

Here $|\cdot|$ is the Euclidean norm and $(\cdot)^T$ denotes transpose.

Diffeomorphic Deformations via Basis Expansions and Elastic Extensions

We construct the map $z \mapsto \chi_z$ utilizing a basis expansion on the reference domain Ω_0 :

$$\chi_z(X) = X + \sum_{i=1}^{d_z} z_i \mathfrak{d}_i(X), \quad (3)$$

where $\{\mathfrak{d}_i\}_{i=1}^{d_z}$ are smooth displacement basis functions $\mathfrak{d}_i : \Omega_0 \rightarrow \mathbb{R}^d$. If

$$\left\| \sum_{i=1}^{d_z} z_i \nabla_X \mathfrak{d}_i \right\|_{L^\infty(\Omega_0)} < 1, \quad (4)$$

then χ_z given by (3) is a diffeomorphism.

However, (4) is often too conservative [2]. In practice, we select $\mathcal{Z}_{ad}$ to be constrained by box bounds $z_l \leq z_i \leq z_u$ that ensures the resulting deformations χ_z are diffeomorphic, and subsequently choose μ_z to be a uniform distribution over $\mathcal{Z}_{ad}$.

Diffeomorphic Deformations via Elastic Extensions

Many shape parametrizations prescribe only the boundary motion $\mathfrak{d}_D^{(i)}$ on the design boundary $\Gamma_D \subset \partial\Omega_0$. The reference-domain formulation, however, requires a volume deformation $\chi_z : \Omega_0 \rightarrow \Omega_z$ defined on all of Ω_0 .

Linear elastic mesh extension: Propagate the prescribed boundary displacement to the interior

$$\begin{aligned} -\nabla \cdot (a_\mu (\nabla \mathfrak{d}_i + \nabla \mathfrak{d}_i^T) + a_\lambda (\nabla \cdot \mathfrak{d}_i) I) &= 0 && \text{in } \Omega_0 \\ \mathfrak{d}_i &= \mathfrak{d}_D^{(i)} && \text{on } \Gamma_D \\ \mathfrak{d}_i &= 0 && \text{on } \partial\Omega_0 \setminus \Gamma_D \end{aligned}$$

The Lamé parameters a_λ, a_μ may be chosen spatially to improve mesh quality and the conditioning of $F_{\chi_z}(X) = \nabla_X \chi_z(X)$.

Remark: Even for free-form deformations with basis functions prescribed throughout Ω_0 , we re-extend their $\mathfrak{d}_i|_{\partial\Omega_0}$ into the interior via linear elasticity. This preserves the induced boundary motion while producing interior basis functions with better mesh quality.

Derivatives of the Solution Operator

By working in the reference domain Ω_0 , Fréchet derivatives $D_m u$ and $D_z u$ can be computed via implicit differentiation **without explicitly involving shape calculus**.

Jacobian-Vector Product (JVP) for input space direction (h_m, h_z) yields h_u :

$$D_u \mathcal{R}(u, m, z) h_u = - (D_m \mathcal{R}(u, m, z) h_m + D_z \mathcal{R}(u, m, z) h_z).$$

Vector-Jacobian Product (VJP) adjoint action for direction $v \in \mathcal{U}$ solves:

$$D_u \mathcal{R}(u, m, z)^T p = -v^T,$$

for the adjoint variable $p \in \mathcal{U}$, such that

$$v^T D_m u(m, z) = D_m \mathcal{R}(u, m, z)^T p, \quad v^T D_z u(m, z) = D_z \mathcal{R}(u, m, z)^T p.$$

Thus, after $u(m, z)$ is computed, each JVP or VJP requires one linearized state or adjoint solve. Multiple derivative actions at the same (m, z) share the same operator $D_u \mathcal{R}$ and differ only in the right hand side.

Shape Optimization under Uncertainty

Let $Q_z : \mathcal{U}(\Omega_z) \times \mathcal{M}(\Omega_z) \rightarrow \mathbb{R}$ denote a scalar QoI on Ω_z . After pullback to Ω_0 ,

$$Q(u, m, z) = Q_z(\tilde{u}, \tilde{m}), \quad u = T_{\chi_z} \tilde{u}, \quad m = T_{\chi_z} \tilde{m}.$$

The uncertain parameter is modeled by a Borel probability measure μ_m on $\mathcal{M} = \mathcal{M}(\Omega_0)$. For each z , the map

$$m \mapsto Q(u(m, z), m, z)$$

is a random variable.

The reference-domain OUU problem is

$$\min_{z \in \mathcal{Z}_{ad}} \mathcal{J}(z) := \rho_{m \sim \mu_m}(Q(u(m, z), m, z)) + \mathcal{P}(z), \quad (5)$$

where $\mathcal{P}$ penalizes or regularizes the shape.

Formulating Risk-Averse Setup for OUU

For a random variable $\xi = Q(u(m, z), m, z)$ with $m \sim \mu_m$:

We utilize risk measures $\rho(\xi)$ to summarize distributions expressing particular risk preferences:

1. **Mean** (Risk-Neutral Baseline):

$$\rho^{\text{mean}}(\xi) = \mathbb{E}[\xi] \quad (6)$$

2. **Entropic Risk Measure** ($\beta > 0$):

$$\rho^{\text{entropic}, \beta}(\xi) = \frac{1}{\beta} \log \mathbb{E}[\exp(\beta \xi)]. \quad (7)$$

3. **Conditional Value-at-Risk (CVaR)** with quantile $\beta \in [0, 1)$:

$$\rho^{\text{CVaR}, \beta}(\xi) = \min_{t \in \mathbb{R}} \left\{ t + \frac{1}{1 - \beta} \mathbb{E}[(\xi - t)^+] \right\} \quad (8)$$

With SAA samples $m^{(i)} \sim \mu_m$, objective and gradient evaluations require state solves $u(m^{(i)}, z)$ and adjoint VJPs for each sample. This sampling cost motivates replacing u by a differentiable operator surrogate u_θ .

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Reduced Basis Neural Operators (RBNO)

Since $\mathcal{M} = \mathcal{M}(\Omega_0)$ and $\mathcal{U} = \mathcal{U}(\Omega_0)$ are separable Hilbert spaces, we can represent the various functions in orthonormal bases $\{\phi_i\}_{i=1}^{\infty}$ of $\mathcal{U}$ and $\{\psi_i\}_{i=1}^{\infty}$ of $\mathcal{M}$, such that for any $u \in \mathcal{U}$ and $m \in \mathcal{M}$,

$$u = \sum_{i=1}^{\infty} \langle u, \phi_i \rangle_{\mathcal{U}} \phi_i \quad \langle \phi_i, \phi_j \rangle_{\mathcal{U}} = \delta_{ij}, \quad (9a)$$

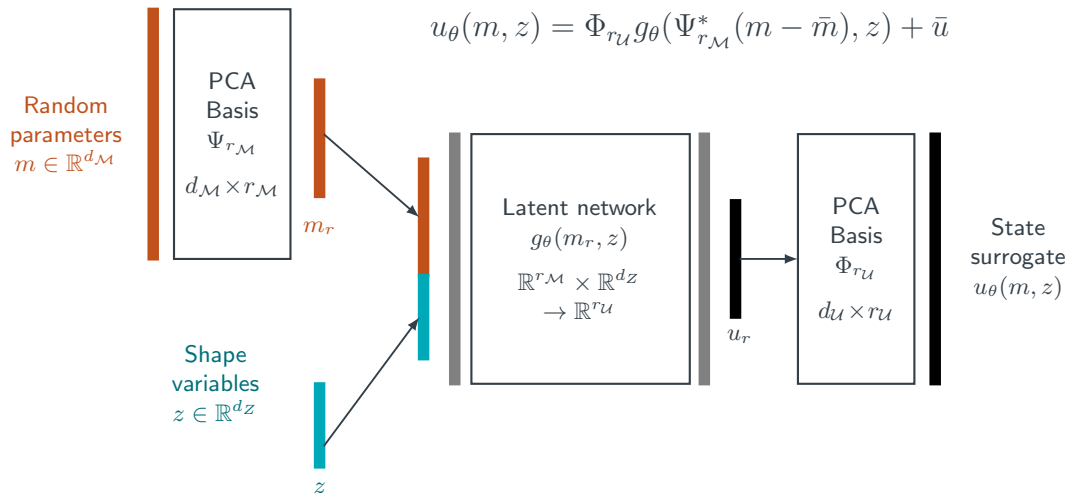
$$m = \sum_{i=1}^{\infty} \langle m, \psi_i \rangle_{\mathcal{M}} \psi_i \quad \langle \psi_i, \psi_j \rangle_{\mathcal{M}} = \delta_{ij}. \quad (9b)$$

Reduced basis architectures represent each of the infinite-dimensional fields via a truncated basis expansion. The corresponding linear encoding and decoding operations

	Encoding	Decoding
$\mathcal{U}$ with basis $\{\phi_i\}_i$	$\mathcal{U} \ni u \mapsto \Phi_{r_u}^* u = (\langle u, \phi_i \rangle_{\mathcal{U}})_{i=1}^{r_u} \in \mathbb{R}^{r_u}$	$\mathbb{R}^{r_u} \ni c \mapsto \Phi_{r_u} c = \sum_{i=1}^{r_u} c_i \phi_i \in \mathcal{U}$
$\mathcal{M}$ with basis $\{\psi_i\}_i$	$\mathcal{M} \ni m \mapsto \Psi_{r_m}^* m = (\langle m, \psi_i \rangle_{\mathcal{M}})_{i=1}^{r_m} \in \mathbb{R}^{r_m}$	$\mathbb{R}^{r_m} \ni c \mapsto \Psi_{r_m} c = \sum_{i=1}^{r_m} c_i \psi_i \in \mathcal{M}$
$\mathcal{Z} = \mathbb{R}^{d_z}$	Identity	Identity

A neural network is then used to represent mapping between the reduced basis coefficients of the inputs and outputs.

Reduced Basis Neural Operators (RBNO)



Shape-DINO Sobolev Objective Loss

Generating full order derivatives and training on the full $\text{HS}(\mathcal{M}, \mathcal{U})$ and $\text{HS}(\mathcal{Z}, \mathcal{U})$ norms lead to prohibitive computational costs. Thus, Shape-DINO trains the RBNO with the latent-space H_μ^1 loss [3, 4]:

$$\begin{aligned} \min_{\theta} \mathbb{E}_{(m,z) \sim \mu} & \left[\|\Phi_{r_u}^*(u(m, z) - b_u) - g_{\theta}(\Psi_{r_m}^*(m - b_m), z)\|_2^2 \right. \\ & + \|\Phi_{r_u}^* D_m u(m, z) \Psi_{r_m} - D_1 g_{\theta}(\Psi_{r_m}^*(m - b_m), z)\|_F^2 \\ & \left. + \|\Phi_{r_u}^* D_z u(m, z) - D_2 g_{\theta}(\Psi_{r_m}^*(m - b_m), z)\|_F^2 \right]. \end{aligned} \quad (10)$$

Here D_1 and D_2 denote derivatives of g_{θ} with respect to its reduced parameter and shape arguments.

This is in contrast to the conventional L_μ^2 training formulation,

$$\min_{\theta} \mathbb{E}_{(m,z) \sim \mu} [\|\Phi_{r_u}^*(u(m, z) - b_u) - g_{\theta}(\Psi_{r_m}^*(m - b_m), z)\|_2^2], \quad (11)$$

which does not involve the derivative error.

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A priori error bounds for surrogate-driven optimization

The operator surrogate $u_\theta(m, z) \approx u(m, z)$ induces a surrogate OUU objective

$$\mathcal{J}_\theta(z) := \rho_{m \sim \mu_m}(Q(u_\theta(m, z), m, z)) + \mathcal{P}(z).$$

Theorem (Optimization Error)

Suppose $\mathcal{B} \subset \mathcal{Z}$ is open and $\mathcal{J} \in C^1(\mathcal{B}; \mathbb{R})$ is λ -strongly convex, i.e.,

$$\mathcal{J}(z_2) \geq \mathcal{J}(z_1) + \langle D_z \mathcal{J}(z_1), z_2 - z_1 \rangle_{\mathcal{Z}' \times \mathcal{Z}} + \frac{\lambda}{2} \|z_2 - z_1\|_{\mathcal{Z}}^2, \quad z_1, z_2 \in \mathcal{B}.$$

Let $\mathcal{Z}_{ad} \subset \mathcal{B}$ be closed, convex, and nonempty, and let $z^\star$ be the unique minimizer of $\mathcal{J}$ over $\mathcal{Z}_{ad}$. If $\mathcal{J}_\theta \in C^1(\mathcal{B}; \mathbb{R})$ approximates $\mathcal{J}$ and $z^\dagger \in \mathcal{Z}_{ad}$ satisfies the first order optimality conditions

$$\langle D_z \mathcal{J}_\theta(z^\dagger), z - z^\dagger \rangle_{\mathcal{Z}' \times \mathcal{Z}} \geq 0, \quad \forall z \in \mathcal{Z}_{ad},$$

then

$$\begin{aligned} \|z^\star - z^\dagger\|_{\mathcal{Z}} &\leq \frac{1}{\lambda} \|D_z \mathcal{J}(z^\dagger) - D_z \mathcal{J}_\theta(z^\dagger)\|_{\mathcal{Z}'}, \\ \mathcal{J}(z^\dagger) - \mathcal{J}(z^\star) &\leq \frac{1}{2\lambda} \|D_z \mathcal{J}(z^\dagger) - D_z \mathcal{J}_\theta(z^\dagger)\|_{\mathcal{Z}'}^2. \end{aligned}$$

Derivative Error for the Expectation

The previous theorem reduces optimization accuracy to controlling $\|D_z \mathcal{J} - D_z \mathcal{J}_\theta\|_{\mathcal{Z}'}$. Now, consider the risk-neutral objective

$$\mathcal{J}^{\text{mean}}(z) := \mathbb{E}_{m \sim \mu_m} [Q(u(m, z), m, z)], \quad \mathcal{J}_\theta^{\text{mean}}(z) := \mathbb{E}_{m \sim \mu_m} [Q(u_\theta(m, z), m, z)],$$

the chain rule gives

$$D_z \mathcal{J}^{\text{mean}}(z) = \mathbb{E}_{m \sim \mu_m} [D_u Q(u(m, z), m, z) D_z u(m, z) + D_z Q(u(m, z), m, z)].$$

Under the paper's moment/Lipschitz assumptions on u , $D_z u$ and C^2 growth assumptions on Q , for every $z \in \mathcal{Z}_{ad}$,

$$\begin{aligned} \|D_z \mathcal{J}^{\text{mean}}(z) - D_z \mathcal{J}_\theta^{\text{mean}}(z)\|_{\mathcal{Z}'} &\leq C_1 \left(\|u - u_\theta\|_{L^2_{\mu_m}(\mathcal{M}; \mathcal{U})} + \|D_z u - D_z u_\theta\|_{L^2_{\mu_m}(\mathcal{M}; \text{HS}(\mathcal{Z}, \mathcal{U}))} \right) \\ &\quad + C_2 \left(\|u - u_\theta\|_{L^2_{\mu_m}(\mathcal{M}; \mathcal{U})}^2 + \|D_z u - D_z u_\theta\|_{L^2_{\mu_m}(\mathcal{M}; \text{HS}(\mathcal{Z}, \mathcal{U}))}^2 \right). \end{aligned}$$

Hence, state accuracy alone is not enough: the objective-gradient error also depends directly on the shape-derivative error, motivating the $D_z u$ term in Shape-DINO training.

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Poisson Problem with Varying Top Boundary

PDE Formulation: We assume that Ω_z is related to $\Omega_0 = (0, 4) \times (0, 1) \subset \mathbb{R}^2$.

$$\begin{aligned} -\nabla_x \cdot (\exp(\tilde{m}) \nabla_x \tilde{u}) &= f && \text{in } \Omega_z \\ \nabla_x \tilde{u} \cdot n &= 0 && \text{on } \Gamma_{L,z} \cup \Gamma_{R,z} \\ \tilde{u} &= 0 && \text{on } \Gamma_{B,z}, \\ \tilde{u} &= 1 && \text{on } \Gamma_{T,z}, \end{aligned}$$

where $x = (x_1, x_2)^T \in \mathbb{R}^2$ and the source term $f(x) = 0.1 \sin(x_1) \cos(x_2)$ is prescribed on Ω_z .

Shape Parametrization of the top boundary (Fourier Basis, $d_z = 11$):

$$x_2(x_1) = a_0 + \sum_{k=1}^5 a_k \cos(k\pi x_1) + b_k \sin(k\pi x_1), \quad z \in [-0.2, 0.2]^{11}.$$

Uncertain Parameter (Log Permeability): Gaussian random field on Ω_0 : $m \sim \mathcal{N}(0, \mathcal{C}_m)$ with $\mathcal{C}_m = (\delta - \gamma \nabla_X \cdot (\Theta \nabla_X))^{-2}$.

Reference Domain Formulation

Function Spaces on Physical Domain:

$$\mathcal{U}(\Omega_z) = \{\tilde{u} \in H^1(\Omega_z) : \tilde{u} = 0 \text{ on } \Gamma_{B,z}, \tilde{u} = 1 \text{ on } \Gamma_{T,z}\},$$

$$\mathcal{V}(\Omega_z) = \{\tilde{v} \in H^1(\Omega_z) : \tilde{v} = 0 \text{ on } \Gamma_{B,z}, \tilde{v} = 0 \text{ on } \Gamma_{T,z}\}.$$

Function Spaces on Reference Domain:

$$\mathcal{U}(\Omega_0) = \{u \in H^1(\Omega_0) : u = 0 \text{ on } \Gamma_B, u = 1 \text{ on } \Gamma_T\},$$

$$\mathcal{V}(\Omega_0) = \{v \in H^1(\Omega_0) : v = 0 \text{ on } \Gamma_B, v = 0 \text{ on } \Gamma_T\}.$$

Weak Pullback Residual:

$$\langle \mathcal{R}(u, m, z), v \rangle = \int_{\Omega_0} \exp(m) (F_{\chi_z}^{-T} \nabla_X u) \cdot (F_{\chi_z}^{-T} \nabla_X v) \det F_{\chi_z} dX - \int_{\Omega_0} (T_{\chi_z} f) v \det F_{\chi_z} dX.$$

Finite Element Space: Piecewise-linear Lagrange basis ($P1$). Dimension $d_m = d_u = 4,257$.

Flux-Tracking Shape Optimization under Uncertainty

Quantity of Interest (tracking objective): The QoI measures the $L^2(\Gamma_{B,z})$ misfit between the normal flux $-e^{\tilde{m}} \nabla_x \tilde{u} \cdot n$ and the target flux τ_{target} :

$$Q_z(\tilde{u}, \tilde{m}) = \frac{1}{2} \int_{\Gamma_{B,z}} (-e^{\tilde{m}} \nabla_x \tilde{u} \cdot n - \tau_{\text{target}})^2 ds \quad (12)$$

QoI in Reference Domain Ω_0 :

$$Q(u, m, z) = \frac{1}{2} \int_{\Gamma_B} (-\exp(m) F_{\chi_z}^{-T} \nabla_X u \cdot N - \tau_{\text{target}})^2 dS \quad (13)$$

Since $\chi_z(X) = X$ on Γ_B , the bottom normal is unchanged in the reference form.

ℓ_2 penalization on z :

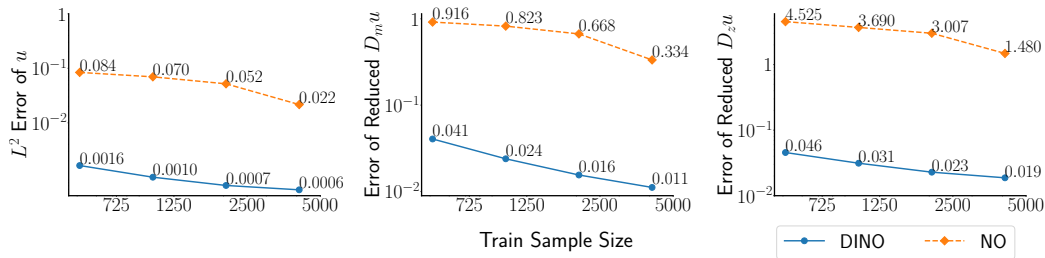
$$\mathcal{P}_{\ell_2}(z; \alpha) := \alpha |\Omega_0| \|z\|_2^2, \quad (14)$$

where $\alpha > 0$ is a penalization weight and the factor $|\Omega_0|$ denotes the Lebesgue measure of Ω_0 .

Poisson Case: Predictive Generalization Accuracy

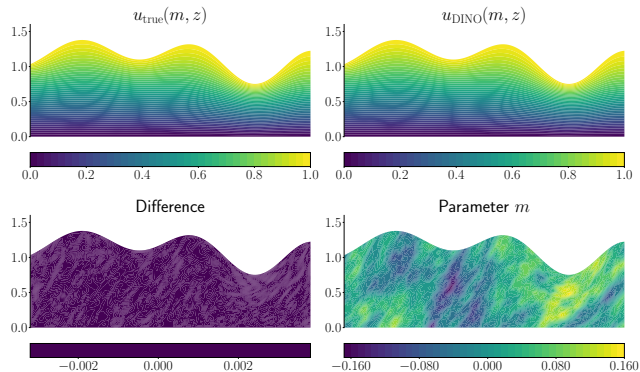
Neural Operator: POD state rank $r_u = 400$, AS parameter rank $r_m = 256$. Architecture: 4×512 width dense net, 2000 epochs.

We plot the mean relative errors in $\mathcal{U} = L^2(\Omega_0)$ of the neural operator output and the mean relative errors of its reduced Jacobians $J_{m,r}(m, z) = \Phi_{r_u}^* D_m u(m, z) \Psi_{r_m}$ and $J_{z,r}(m, z) = \Phi_{r_u}^* D_z u(m, z)$ over a test set of 1,024 samples from the input distribution and 10 different runs of the training process.



Poisson Results: Predictive Capabilities

Example of state u by PDE and shape-DINO plotted on deformed domain.



Visually, Shape-DINO prediction matches the true PDE state.

Poisson Results: Shape Optimization with Mean Objective

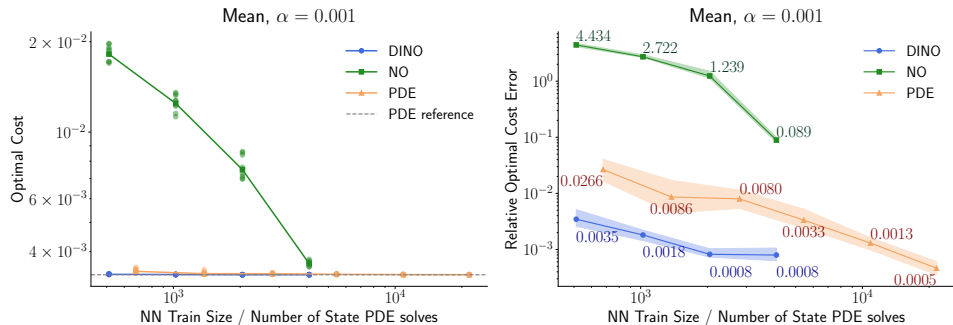


Figure: Risk-Neutral: Expected Value Optimization.

Even when the objective function is easy to estimate, Shape-DINO converges rapidly to the reference solution with substantially fewer PDE solves than the PDE-based method, while Shape-NO exhibits larger errors and slower convergence.

Poisson Results: Shape Optimization with CVaR Objective

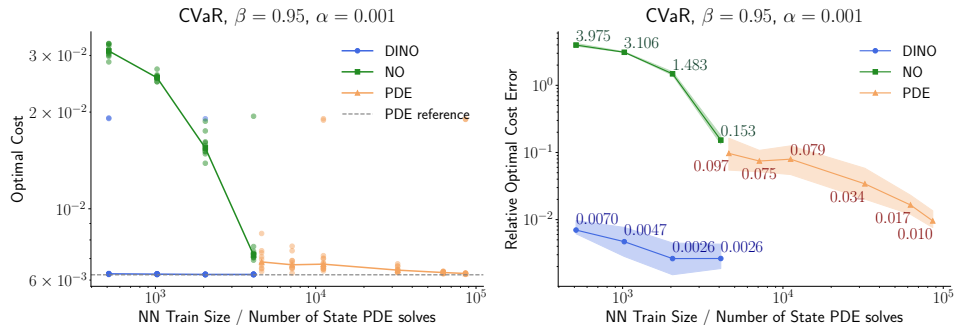


Figure: Risk-Averse: CVaR ($\beta = 0.95$) with small penalty $\alpha = 0.001$.

In this highly risk-sensitive setting, Shape-DINO achieves near-reference accuracy with orders of magnitude fewer PDE solves than the PDE-based SAA, whereas Shape-NO performs poorly for small and moderate training sizes.

Poisson Results: Increased Regularization

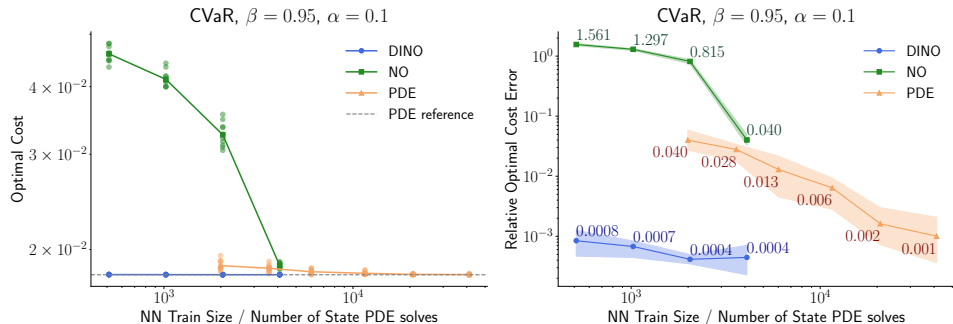


Figure: Risk-Averse: CVaR ($\beta = 0.95$) with larger penalty weight $\alpha = 0.1$.

With the larger shape penalty, the PDE-based SAA and Shape-DINO errors are closer; the derivative-informed surrogate remains competitive at substantially smaller offline PDE solves.

Poisson Results: Flux Alignment

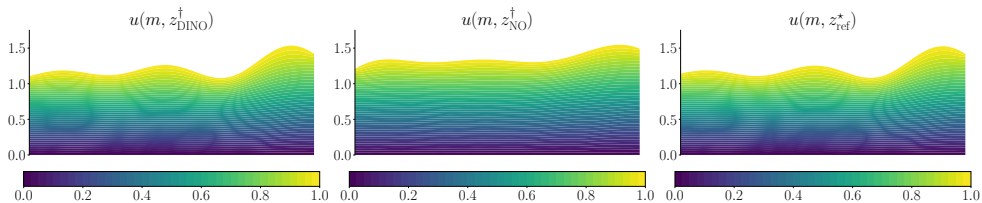


Figure: Optimal Design Geometries.

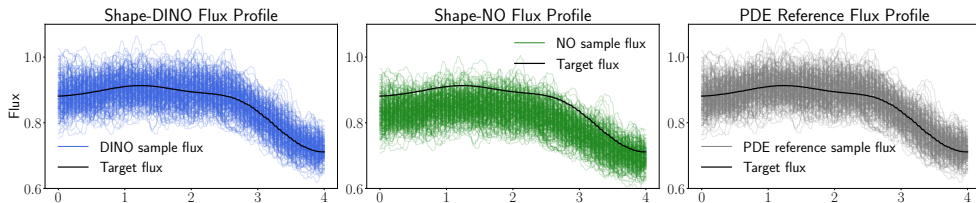
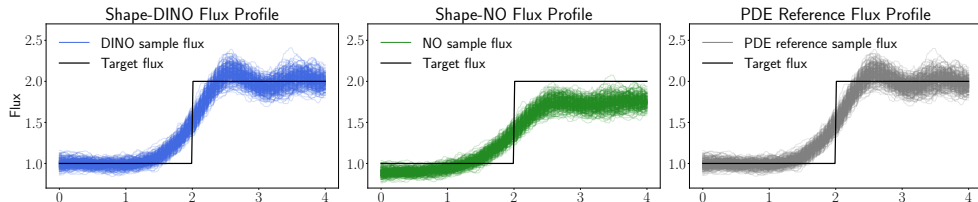


Figure: Smooth Bump Target Profile.

Poisson Generalization: Out-of-Distribution Flux Target

Sample flux profiles at the optimal designs obtained from CVaR optimization ($\beta = 0.95$, $\alpha = 0.001$) for a discontinuous target $\tau_{\text{target}}(x_1) = 1$ on $[0, 2]$ and $\tau_{\text{target}}(x_1) = 2$ on $[2, 4]$.



This target lies outside the training distribution and is not achievable by shapes in the admissible set. Shape-DINO still produces flux profiles that better match the target and remain stable across uncertainty, while Shape-NO significantly deviates from the target.

Dissipation Reduction in 2D Flow under Multiple Uncertain Inflows

Consider a 2D channel $\Omega_{\text{channel}} = (0, 2) \times (0, 1) \subset \mathbb{R}^2$ containing an initially circular object of radius 0.1 centered at $(0.5, 0.5)^T$.

We introduce two flow scenarios $s \in \{\text{East, North}\}$ with

$$\Omega_z^s = \Omega_{\text{channel}} \setminus \overline{\Omega_{\text{object}}^s}.$$

The eastward object is $\Omega_{\text{object}}^{\text{East}} = \chi_z(\Omega_{\text{object}})$; the northward object is the same deformed shape reflected about $x_1 = x_2$.

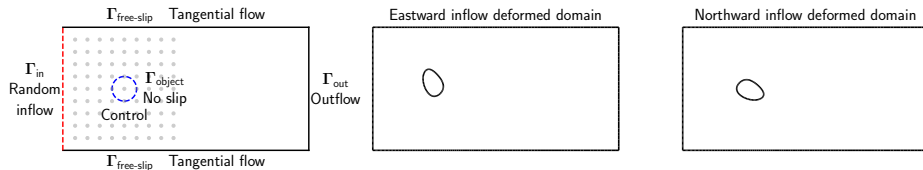


Figure: Reference Domain, Eastward deformation & Northward deformation.

Equivalently, the northward case uses mirrored displacement bases, so both scenarios share the same reference domain while presenting different object orientations to the inflow.

2D NS Spatial PDE

For each scenario $s \in \{\text{East}, \text{North}\}$, the spatial velocity-pressure pair is $\tilde{u}^s = (\tilde{v}^s, \tilde{p}^s)$ on Ω_z^s .

$$(\tilde{v}^s \cdot \nabla_x) \tilde{v}^s - \nabla_x \cdot \tilde{\mathbf{T}}(\tilde{u}^s) = 0 \quad \text{in } \Omega_z^s$$

$$\nabla_x \cdot \tilde{v}^s = 0 \quad \text{in } \Omega_z^s$$

$$\tilde{v}^s = (\tilde{v})_{\text{in}}^s \quad \text{on } \Gamma_{\text{in}}$$

$$\tilde{\mathbf{T}}(\tilde{u}^s)n = 0 \quad \text{on } \Gamma_{\text{out}}$$

$$\tilde{v}^s = 0 \quad \text{on } \Gamma_{\text{no-slip}}$$

$$\tilde{\mathbf{T}}(\tilde{u}^s)n \cdot e_t = 0 \quad \text{on } \Gamma_{\text{free-slip}},$$

$$\tilde{v}^s \cdot n = 0 \quad \text{on } \Gamma_{\text{free-slip}},$$

where

$$\Gamma_{\text{in}} = \{x \in \partial\Omega_z^s : x_1 = 0\}, \quad \Gamma_{\text{no-slip}} = \partial\Omega_{\text{object}}^s,$$

$$\Gamma_{\text{out}} = \{x \in \partial\Omega_z^s : x_1 = 2\}, \quad \Gamma_{\text{free-slip}} = \{x \in \partial\Omega_z^s : x_2 = 0 \text{ or } x_2 = 1\}.$$

2D NS Spatial PDE

The stress tensor $\tilde{\mathbf{T}}$ and strain tensor $\tilde{\varepsilon}$ are defined as

$$\tilde{\mathbf{T}}(\tilde{u}^s) = 2\nu\tilde{\varepsilon}(\tilde{v}^s) - \tilde{p}^s I, \quad \tilde{\varepsilon}(\tilde{v}^s) = \frac{\nabla_x \tilde{v}^s + (\nabla_x \tilde{v}^s)^T}{2}.$$

The inflow profile is determined by m through

$$(\tilde{v})_{\text{in}}^s(x) = \exp(\tilde{m}^s(x))e_1, \quad e_1 = (1, 0)^T.$$

The random inflow field is sampled on the reference domain $\Omega_0 = \Omega_{\text{channel}} \setminus \overline{\Omega_{\text{object}}}$:

$$m^s \sim \mu_m^s = \mathcal{N}(0, \mathcal{C}_m^s), \quad \tilde{m}^s = T_{\chi_z}^{-1} m^s.$$

Each $\mathcal{C}_m^s$ is a Bi-Laplacian covariance operator with correlation length 0.25; the pointwise variance is 0.2 for $s = \text{East}$ and 0.1 for $s = \text{North}$.

2D NS Case: Bernstein FFD Parametrization

Let $\Omega_{\text{control}} = (0, 1)^2$ contain the object and let $\xi_{k,l} = (k/K, l/L)^T$ be a $K \times L$ control lattice. Based on [5],

$$b_j^{n_p}(t) = \binom{n_p}{j} t^j (1-t)^{n_p-j}, \quad b_{k,l}^{K,L}(t_1, t_2) = b_k^K(t_1) b_l^L(t_2).$$

If $\delta_{k,l} \in \mathbb{R}^2$ is the control-point displacement, then

$$\chi_z(X) = X + \sum_{k=1}^{K-1} \sum_{l=1}^{L-1} b_{k,l}^{K,L}(X_1, X_2) \delta_{k,l}.$$

Equivalently, this matches the basis-expansion form with

$$\mathfrak{d}_{k,l}^1(X) = b_{k,l}^{K,L}(X_1, X_2) e_1, \quad \mathfrak{d}_{k,l}^2(X) = b_{k,l}^{K,L}(X_1, X_2) e_2.$$

In the experiments, $K = L = 10$, boundary control points are fixed to keep the outer channel boundary stationary, and the remaining bases are preprocessed by linear elasticity to improve mesh quality.

2D NS Case: Reference-Domain Implementation

The PDE residual is assembled on the fixed reference domain Ω_0 . Because the FFD keeps the outer channel boundary fixed, the left inflow boundary is unchanged under χ_z ; in particular, the boundary normal appearing in the inflow terms does not change.

Under this representation, the inflow data have the simple reference-domain form

$$v_{\text{in}}^s(X) = \exp(m^s(X))e_1, \quad e_1 = (1, 0)^T.$$

Nitsche terms are used to weakly impose this uncertain Dirichlet inflow condition in the variational residual.

Discretization and dimensions

- Taylor–Hood state discretization: v^s continuous piecewise-quadratic, p^s continuous piecewise-linear.
- Random inflow field m^s : continuous piecewise-quadratic, $d_m = 201$ for each scenario.
- Shape variable: $d_z = 162$; Total state dimension: $d_u = 86,205$.

2D NS Dissipation-reducing under Uncertain Inflows

For each inflow scenario $s \in \{\text{East}, \text{North}\}$, the state is $\tilde{u}^s = (\tilde{v}^s, \tilde{p}^s)$ and the Qol is the viscous dissipation rate:

$$Q_z^s(\tilde{u}^s) := \int_{\Omega_z^s} 2\nu \tilde{\varepsilon}(\tilde{v}^s) : \tilde{\varepsilon}(\tilde{v}^s) dx. \quad (15)$$

Pulling the Qol back to Ω_0 gives

$$Q(u^s, m^s, z) = \int_{\Omega_0} 2\nu \varepsilon(v^s) : \varepsilon(v^s) \det F dX. \quad (16)$$

The risk objective for each scenario is

$$\mathcal{J}^s(z) := \rho_{m^s \sim \mu_{m^s}}^{\text{entropic}, \beta}(Q(u^s(m^s, z), m^s, z)), \quad s \in \{\text{East}, \text{North}\}.$$

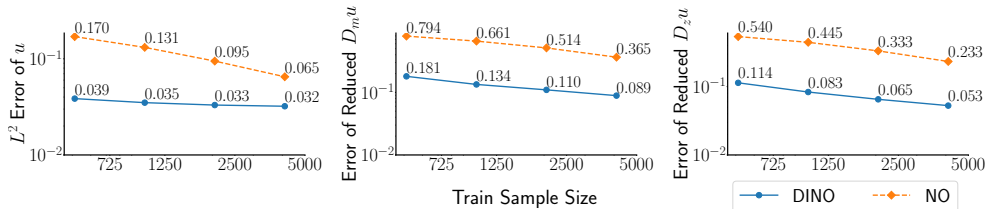
The multi-scenario OUU problem combines the two scenario risks:

$$\min_{z \in \mathcal{Z}_{ad}} \mathcal{J}^{\text{weighted}}(z), \quad \mathcal{J}^{\text{weighted}}(z) = \mathcal{J}^{\text{East}}(z) + 0.3 \mathcal{J}^{\text{North}}(z) + \mathcal{P}_{\ell_2}(z; \alpha). \quad (17)$$

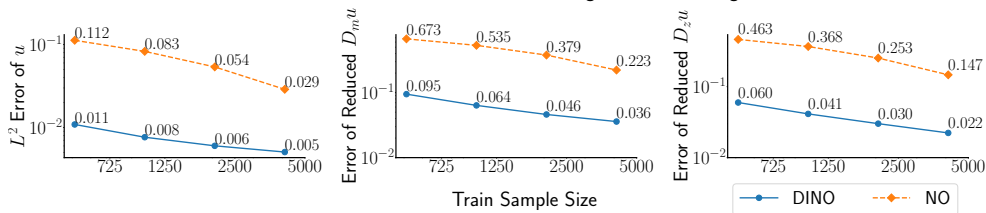
Here $\beta = 1$, $\alpha = 0.001$, and $\mathcal{Z}_{ad} = \{z \in [-1, 1]^{d_z} : \text{Vol}(z) \in [0.3V_0, 1.2V_0]\}$. Two neural operators are trained independently, one for each scenario.

2D NS Case: Predictive Generalization Accuracies

Eastward Inflow Neural Network Average Relative Testing Error



Northward Inflow Neural Network Average Relative Testing Error



2D NS Results: Flow Fields at Initial and Optimal Designs

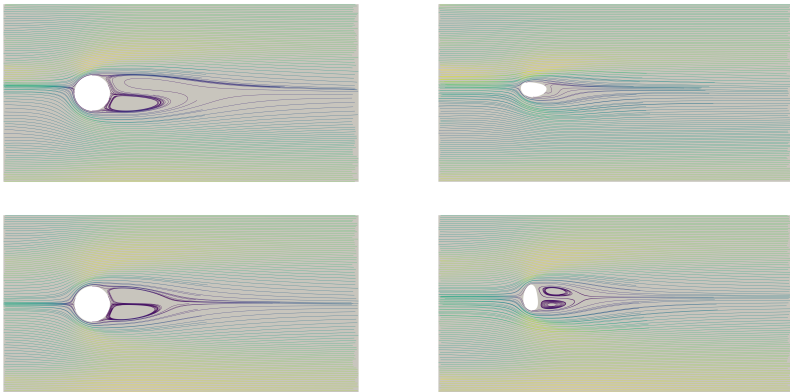


Figure: Velocity fields comparison: Uncontrolled (Left) vs DINO-Optimal (Right) for East/North flows.

The optimized geometry $z_{\text{DINO}}^\dagger$, though not symmetric, significantly reduces recirculation behind the object and lowers the overall viscous dissipation rate compared to the initial shape.

2D NS Results: Entropic Risk Objective Comparison

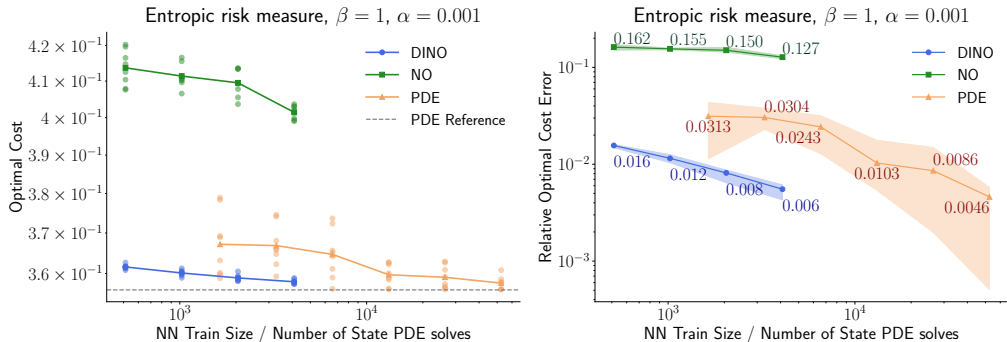


Figure: Cost comparison against direct PDE optimization.

Note that the counts reported above refer to the number of samples per flow scenario. The actual number of PDE solves is twice the reported number since we evaluate the QoI for both the eastward flow and northward flow scenarios.

3D NS Case: High-Fidelity Channel Tower Flow

Geometry: flow past a frustum-shaped tower Ω_{object} in

$$\Omega_{\text{channel}} = (0, 2) \times (0, 1) \times (0, 1) \subset \mathbb{R}^3.$$

For each flow scenario,

$$\Omega_z^s = \Omega_{\text{channel}} \setminus \overline{\Omega_{\text{object}}^s}, \quad s \in \{\text{East}, \text{North}\}.$$

Channel faces and roles

$\Gamma_{\text{left}} : x_1 = 0$	$\Gamma_{\text{right}} : x_1 = 2$	inflow/outflow
$\Gamma_{\text{base}} : x_2 = 0$	$\Gamma_{\text{top}} : x_2 = 1$	ground/free-slip
$\Gamma_{\text{front}} : x_3 = 0$	$\Gamma_{\text{back}} : x_3 = 1$	free-slip

No-slip is imposed on $\Gamma_{\text{base}} \cup \Gamma_{\text{object}}^s$.

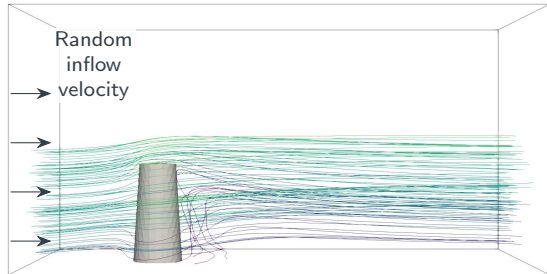


Figure: Reference tower with streamlines.

3D NS Case: Scenarios and Random Inflow

The same steady incompressible Navier–Stokes formulation from the 2D case is used, with 3D velocity–pressure state $\tilde{u}^s = (\tilde{v}^s, \tilde{p}^s)$ and

$$\tilde{\mathbf{T}}(\tilde{u}^s) = 2\nu\tilde{\varepsilon}(\tilde{v}^s) - \tilde{p}^s I, \quad \tilde{\varepsilon}(\tilde{v}^s) = \frac{\nabla_x \tilde{v}^s + (\nabla_x \tilde{v}^s)^T}{2}.$$

Scenario geometry: the eastward object is $\Omega_{\text{object}}^{\text{East}} = \chi_z(\Omega_{\text{object}})$. The northward object is obtained by reflecting the eastward geometry about the plane $X_1 = X_3$.

Boundary assignment

$$\Gamma_{\text{in}} = \Gamma_{\text{left}}, \quad \Gamma_{\text{out}} = \Gamma_{\text{right}}, \quad \Gamma_{\text{no-slip}} = \Gamma_{\text{base}} \cup \Gamma_{\text{object}}^s, \quad \Gamma_{\text{free-slip}} = \Gamma_{\text{front}} \cup \Gamma_{\text{top}} \cup \Gamma_{\text{back}}.$$

Random inflow: unlike the 2D case, the inflow scales an atmospheric boundary-layer power law,

$$(\tilde{v}^s)_{\text{in}}(x) = \exp(\tilde{m}^s(x)) x_2^{1/5} e_1, \quad e_1 = (1, 0, 0)^T,$$

with Gaussian random field sampled on the reference domain,

$$\tilde{m}^s = T_{\chi_z}^{-1} m^s, \quad m^s \sim \mathcal{N}(0, \mathcal{C}_m^s).$$

The reference-domain residual is discretized on a tetrahedral mesh with Taylor–Hood elements for (v, p) and quadratic elements for m , giving $d_u = 393,331$ and $d_m = 3,593$.

3D NS Case: Shape Definition

Shape parametrization: Bernstein FFD with a $6 \times 6 \times 6$ control lattice. Control points lie at $\{0, 0.2, 0.4, 0.6, 0.8, 1\}$ along each axis. Points on the fixed channel faces are excluded, and allowed motions are only in the horizontal X_1 – X_3 plane.

Thus 80 movable control points contribute two horizontal degrees of freedom each, giving $d_z = 160$ while keeping the tower height fixed.

Two inflow scenarios: $s \in \{\text{East, North}\}$. The eastward object is $\Omega_{\text{object}}^{\text{East}} = \chi_z(\Omega_{\text{object}})$, while the northward displacement basis is obtained by reflecting about the plane $X_1 = X_3$:

$$(X_1, X_2, X_3) \mapsto (X_3, X_2, X_1), \quad (e_1, e_2, e_3) \mapsto (e_3, e_2, e_1).$$

Reference-domain parametrization: both the PDE and the neural operator are formulated on Ω_0 through z , so geometry changes enter through the transformed residual.

3D NS Case: Force and Moment Objectives

For each scenario $s \in \{\text{East}, \text{North}\}$, the hydrodynamic force and bending moment on the tower boundary are

$$\mathbf{F}_z^s(\tilde{u}^s, \tilde{u}^s) = \int_{\Gamma_{\text{object}}^s} \tilde{\mathbf{T}}(\tilde{u}^s) n \, ds,$$
$$\mathbf{M}_z^s(\tilde{u}^s, \tilde{m}^s) = \int_{\Gamma_{\text{object}}^s} (x - O) \times (\tilde{\mathbf{T}}(\tilde{u}^s) n) \, ds.$$

Here $O = (0.5, 0, 0.5)^T$ is the center of the tower base, and the force and moment vectors are expressed in spatial coordinates.

Writing $\mathbf{F}^s = (F_1^s, F_2^s, F_3^s)^T$ and $\mathbf{M}^s = (M_1^s, M_2^s, M_3^s)^T$, the two optimization Qols are

$$Q_{\text{force}}^s(u^s, m^s, z) := (F_1^s)^2 + (F_3^s)^2, \quad \text{horizontal force,}$$
$$Q_{\text{bend}}^s(u^s, m^s, z) := \|\mathbf{M}^s(u^s, m^s, z)\|_2^2 = (M_1^s)^2 + (M_2^s)^2 + (M_3^s)^2, \quad \text{bending moment.}$$

Both Qols are defined per scenario and are combined through weighted risk objectives across inflows.

3D NS Case: Risk Objectives and Penalization

For each scenario, the entropic-risk objectives are

$$\begin{aligned}\mathcal{J}_{\text{force}}^s(z) &:= \rho_{m^s \sim \mu_m^s}^{\text{entropic}, \beta}(Q_{\text{force}}^s(u^s(m^s, z), m^s, z)), \\ \mathcal{J}_{\text{bend}}^s(z) &:= \rho_{m^s \sim \mu_m^s}^{\text{entropic}, \beta}(Q_{\text{bend}}^s(u^s(m^s, z), m^s, z)).\end{aligned}$$

The two multi-scenario OUU objectives use different regularization terms:

$$\begin{aligned}\mathcal{J}_{\text{force}}(z) &= 0.75 \mathcal{J}_{\text{force}}^{\text{East}}(z) + 0.25 \mathcal{J}_{\text{force}}^{\text{North}}(z) + \mathcal{P}_{\ell_2}(z; \alpha), \quad \alpha = 2 \times 10^{-5}, \\ \mathcal{J}_{\text{bend}}(z) &= 0.75 \mathcal{J}_{\text{bend}}^{\text{East}}(z) + 0.25 \mathcal{J}_{\text{bend}}^{\text{North}}(z) + \mathcal{P}_{\text{grav}}(z; \alpha), \quad \alpha = 1.\end{aligned}$$

Shape-size penalty

$$\mathcal{P}_{\ell_2}(z; \alpha) := \alpha |\Omega_0| \|z\|_2^2.$$

This penalizes large FFD coefficient magnitudes.

Gravity-moment penalty

$$\begin{aligned}\mathbf{M}^g(z) &:= \int_{\Omega_{\text{object}}} (x - O) \times \rho_{\text{object}} g e_2 \, dx, \\ \mathcal{P}_{\text{grav}}(z; \alpha) &:= \frac{\alpha}{2} \|\mathbf{M}^g(z)\|_2^2.\end{aligned}$$

Here $\rho_{\text{object}} = 2000$, $g = 9.81$, and $e_2 = (0, -1, 0)^T$.

3D NS Case: Neural Operator Training Setup

Training samples are generated from the same input distributions used in the OUU problem:

$$z \sim \text{Unif}([-0.5, 0.5]^{d_z}), \quad m^s \sim \mu_m^s, \quad s \in \{\text{East}, \text{North}\}.$$

Because each 3D Navier–Stokes solve is expensive, both Shape-DINO and Shape-NO are trained with only 384 samples. The reduced spaces use

$$n_{\text{POD}} = n_{\text{AS}} = 256, \quad r_{\mathcal{U}} = 200, \quad r_{\mathcal{M}} = 128.$$

The latent map is a two-hidden-layer dense network of width 512, trained with Adam for 2000 epochs. On 128 test samples, Shape-DINO/Shape-NO relative errors are 7.22%/18.91% for eastward flow and 4.13%/12.55% for northward flow.

The same trained networks are deployed for both horizontal-force and bending-moment OUU. Since a PDE-based SAA reference is infeasible in 3D, the PDE baseline is deterministic at $m^s = \bar{m}^s$, while the neural-operator optimizations use SAA with 1024 samples per iteration.

3D NS Results: Bending Moment Distribution under Test Inflows

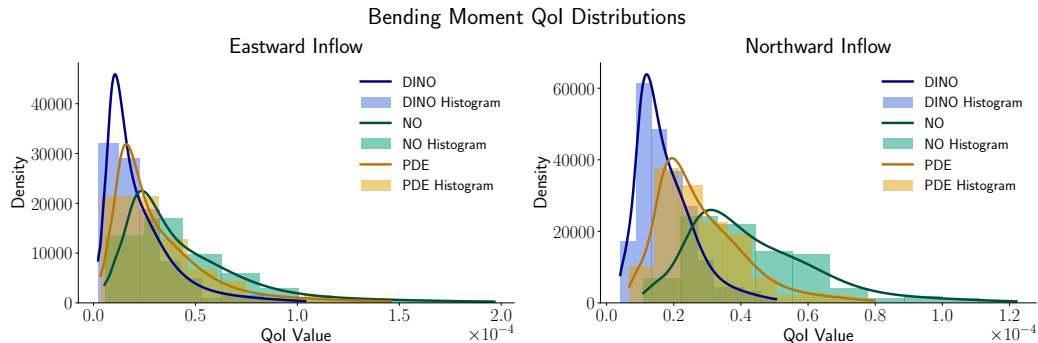


Figure: Bending Moment QoI Distribution

Relative to the deterministic PDE and Shape-NO designs, the Shape-DINO design shifts the bending-moment distribution closer to zero and reduces its spread under the test inflow ensemble.

3D NS Results: Horizontal Distribution under Test Inflows

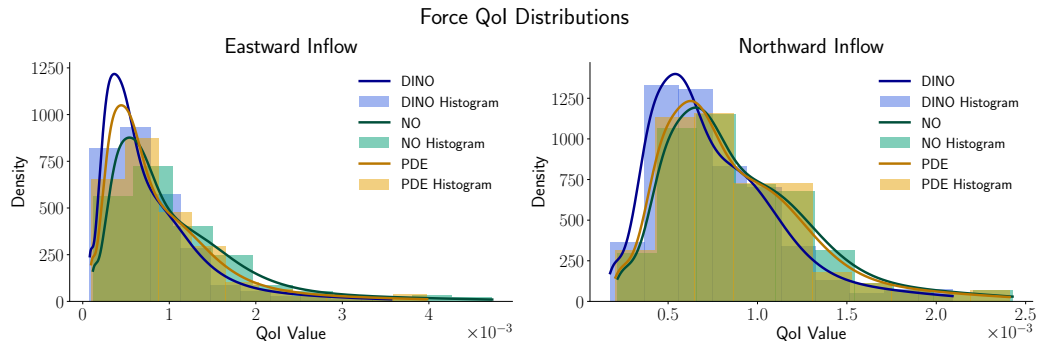


Figure: Horizontal Total Force Qol Distribution

Shape-DINO yields designs that are simultaneously robust with respect to multiple performance metrics, reducing both the magnitude and variability of the responses under uncertainty.

3D NS Results: Optimal Geometric Streamlines

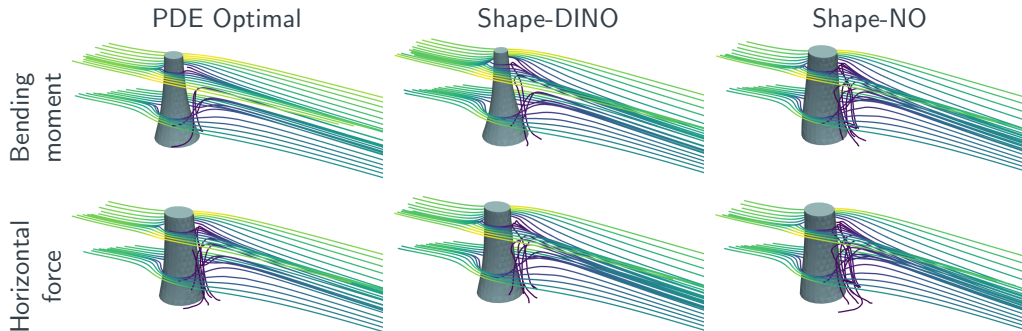


Figure: Optimal designs for the two 3D Qols, evaluated under eastward mean inflow.

Consistent with the Qol distributions, Shape-DINO produces thinner designs with smoother flow passage, while Shape-NO yields bulkier geometries with stronger flow deflection and larger downstream wakes.

Architecture and Computational Speedups

Network/training configuration

Example	$d_{W,i}$	Initial LR	LR scheduler, $\gamma = 0.5$	Epochs	r_u	r_m	d_z
Poisson	[512, 512, 512, 512]	5×10^{-4}	{800, 1200, 1800}	2000	400	256	11
2D NS	[512, 1024, 512]	10^{-4}	{500, 750}	1000	256	50	162
3D NS	[512, 512]	10^{-4}	{1000, 1500}	2000	200	128	160

Online evaluation time (seconds per sample)

Problem	State Solve			VJP / Adjoint		
	PDE	NN (1)	NN (1000)	PDE	NN (1)	NN (1000)
Poisson	1.06e-1	4.12e-4	5.50e-7	8.84e-2	1.25e-5	1.35e-6
2D NS	5.08e+0	3.86e-4	1.70e-6	1.07e+1	1.15e-4	3.16e-6
3D NS	7.18e+2	3.85e-4	5.46e-6	1.62e+2	9.68e-4	1.07e-5

In 3D NS, batched neural-operator evaluations reduce both state and VJP costs by many orders of magnitude relative to high-fidelity PDE/adjoint solves.

Conclusions

- Shape-DINO is a reference-domain derivative-informed neural operator framework for PDE-constrained shape OUU, designed to approximate both $u(m, z)$ and the sensitivities needed by optimization.
- Incorporating Jacobian information improves state prediction, gradient fidelity, and QoI reliability compared with standard L^2_μ neural-operator training.
- Shape-DINO improves sampling efficiency: the offline PDE solves used for training are amortized across many SAA samples, different objectives, and risks choices.
- The resulting online evaluations and VJPs are orders of magnitude faster than high-fidelity PDE/adjoint solves, enabling large-batch risk estimation in settings where direct PDE-based SAA is impractical.

Limitations and Further Directions

- Shape-DINO is most useful when uncertainty, geometry variation, and repeated optimization queries make PDE-based SAA expensive; for simple objectives, direct PDE optimization may be preferable.
- The current framework relies on diffeomorphic mesh deformation and elastic extensions, so the admissible shape family must avoid severe mesh distortion.
- Performance depends on the training distribution: extrapolating far outside the sampled uncertainty or shape regime can degrade both state and derivative accuracy.
- The paper also gives supporting theory not emphasized in this talk, including reduced-basis reconstruction error estimates and universal approximation results for the multi-input RBNO in suitable C^1 norms.
- Natural extensions include adaptive sampling during optimization, higher-order derivative information, and richer shape parameterizations.

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Questions?

Thank You!